

Discussion of
“The Geography of Fuel Tax Incidence”
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What this paper does

- Cleverly estimates a structural Logit/Box-Cox demand for diesel *without quantity data* using marginal costs and supply side optimality conditions.
- Measures heterogeneous **welfare incidence** of a carbon–fuel tax across 2,220 French local markets.
- Welfare impacts of counterfactual carbon tax schemes

Demand + Supply – core equations

Individual i in market ℓ choosing station j :

$$U_{ij}^{\ell} = X_j^{\ell'} \beta_{\ell} + \alpha_{\ell} p_j^{\ell} + \xi_j^{\ell} + \varepsilon_{ij}^{\ell}$$

Logit share:

$$s_j^{\ell}(p) = \frac{\exp(X_j^{\ell'} \beta_{\ell} + \alpha_{\ell} p_j^{\ell} + \xi_j^{\ell})}{1 + \sum_{k=1}^{J_{\ell}} \exp(X_k^{\ell'} \beta_{\ell} + \alpha_{\ell} p_k^{\ell} + \xi_k^{\ell})}.$$

Bertrand FOC with single-product firms yields:

$$s_j^{\ell} = 1 + \frac{1}{\alpha_{\ell}(p_j^{\ell} - c_j^{\ell})}$$

Outside-good share:

$$s_0^{\ell} = 1 - \sum_{j=1}^{J_{\ell}} s_j^{\ell} = 1 - J_{\ell} - \sum_{j=1}^{J_{\ell}} \frac{1}{\alpha_{\ell}(p_j^{\ell} - c_j^{\ell})}.$$

Deriving $\xi_j^\ell(\alpha_\ell, \beta_\ell)$

Logit inversion gives

$$\xi_j^\ell = \log s_j^\ell - \log s_0^\ell - X_j^{\ell'} \beta_\ell - \alpha_\ell p_j^\ell.$$

Substitute shares:

$$\xi_j^\ell(\alpha_\ell, \beta_\ell) = \log\left(1 + \frac{1}{\alpha_\ell}(p_j^\ell - c_j^\ell)\right) - \log\left(1 - J_\ell - \sum_{k=1}^{J_\ell} \frac{1}{\alpha_\ell}(p_k^\ell - c_k^\ell)\right) - X_j^{\ell'} \beta_\ell - \alpha_\ell p_j^\ell. \quad (3)$$

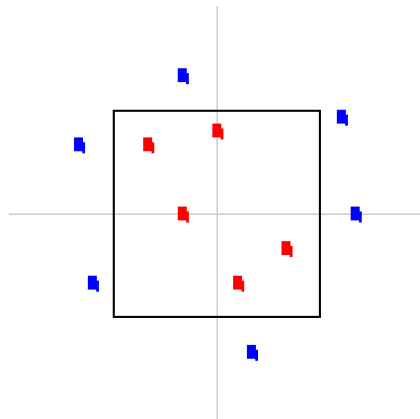
Concentrating out β_ℓ expresses ξ as function of prices, costs, and α^ℓ :

$$\beta_\ell(\alpha_\ell) = (X^{\ell'} X^\ell)^{-1} X^{\ell'} [\log s^\ell(\alpha_\ell) - \log s_0^\ell(\alpha_\ell) - \alpha_\ell p^\ell].$$

With ξ_j^ℓ , use BLP-like objective despite no quantity data.

Spatial delineation of markets

- Hierarchical clustering (≤ 15 stations, 5–10 km radii).
 - Peripheral stations likely compete across boundaries.
 - Cross-border competitive pressure $\rightarrow$ margins, pass-through.
- Possible idea. When computing ξ_j^ℓ for one gas station, create custom market centered at it. Repeat for each gas station.



- **Supply-side inversion** powerful but relies on *static* profit maximisation.
 - Evidence of persistent price deviations ([Dubé and MisraDubé and Misra2023]).
- Requires accurate c_j^ℓ .
- 12.5% of markets with unsatisfied share constraints raises concerns.
- Extension: joint-profit ownership matrix for gas chains?

Why a “No-Quantities” Estimation Strategy is Appealing

- **Dramatically lighter data requirements**
 - Station-level **prices** are often public and easy to scrape.
 - Bypasses the hardest variable to obtain (reliable **quantities**)
- **Needed marginal-cost information can be remarkably transparent (in digital markets)**
 - For purely digital goods the incremental cost is essentially $c_j \approx 0$.
 - When license fees exist they are often publicly stated: e.g. Apple App Store / iTunes take 30%.

Counterfactuals – equity & dynamics

- Within-market heterogeneity (income, commute distance) assumed away $\Rightarrow$ tax that looks progressive may hide **intra-market regressivity**.
- Short-run focus: durable responses (vehicle choice, relocation, station exit) matter for long-run incidence.
- Figure 7 explores 56k income–tax polynomials — could instead *optimize*:
 - Equalise CS loss across incomes; or maximise median-voter utility.
- Communication to voters: can they understand a 4th-degree tax schedule?

Takeaways

- Elegant method to recover demand & incidence *without quantities*.
- Opens up many possible research projects (big contribution)
- Possible extensions



Dubé, J.-P. and S. Misra (2023): “Personalized Pricing and Consumer Welfare,” *Journal of Political Economy*, 131, 131–189.